

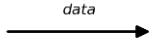
## Data Flow Diagram

|                      |               |
|----------------------|---------------|
| <b>Date</b>          | 15-07-2026    |
| <b>Team ID</b>       |               |
| <b>Project Name</b>  | Rising Waters |
| <b>Maximum Marks</b> | 2 Marks       |

### Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

### DFD Symbol Legend

| Symbol                                                                              | Name                   | Description                                                                                                          |
|-------------------------------------------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------|
|    | <b>External Entity</b> | A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway). |
|  | <b>Process</b>         | An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).                |
|  | <b>Data Store</b>      | A place where data is held for later use (e.g., Customer Database, Inventory).                                       |
|  | <b>Data Flow</b>       | The movement of data between entities, processes, and data stores, labeled with what data is being passed.           |

### External Entities

- User
- Machine Learning Model

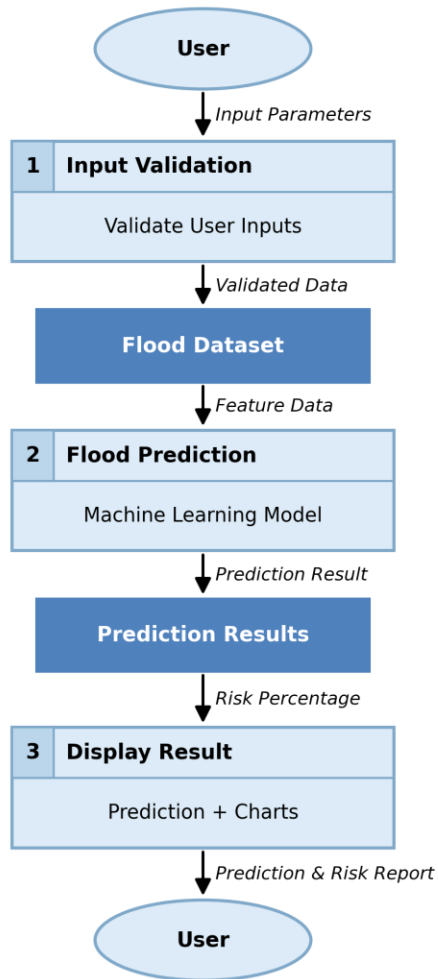
### Processes

- 1. Input Validation
  - Validate user inputs such as rainfall, latitude, longitude, elevation, slope, etc.
- 2. Flood Prediction
  - Process the validated data using the trained Machine Learning model.
- 3. Display Result
  - Show flood prediction, flood risk percentage, and graphical visualization.

### Data Stores

- Flood Dataset
- Prediction Results

### Data Flow



### Labels for the Arrows

- User → Input Validation: Input Parameters
- Input Validation → Flood Dataset: Validated Data
- Flood Dataset → Flood Prediction: Feature Data
- Flood Prediction → Prediction Results: Prediction Result
- Prediction Results → Display Result: Risk Percentage
- Display Result → User: Flood Prediction, Risk %, Charts